Database Design — Touiteur

1. Elementary Data Table (Data Dictionary)

This dictionary lists all the elementary attributes required to model the *Touiteur* application in Oracle Database.

Attribute Code	Description	Oracle Type	Domain	Rules and Constraints
`id_user`	Unique identifier of a registered user	**`NUMBER(10)`**	User	Primary key, mandatory
`nom_user`	User's last name	**`VARCHAR2(50)`**	User	Mandatory (**`NOT NULL`**)
`prenom_user`	User's first name	**`VARCHAR2(50)`**	User	Mandatory (**`NOT NULL`**)
`email_user`	Email address of the user profile	**`VARCHAR2(100)`**	User	Mandatory (**`NOT NULL`**)
`nb_touites`	Number of touites created by the user	**`NUMBER(10)`**	User	Mandatory, default **`0`**, value ≥ 0, updated via trigger
`id_touite`	Unique identifier of a touite	**`NUMBER(10)`**	Touite	Primary key, mandatory
`texte_touite`	Text content of the published touite	**`VARCHAR2(235)`**	Touite	Mandatory, limited to 235 characters
`date_pub`	Publication date of the touite	**`DATE`**	Touite	Mandatory, default **`SYSDATE`**
`id_tag`	Unique identifier of a tag	**`NUMBER(10)`**	Tag	Primary key, mandatory
`libelle_tag`	Tag label	**`VARCHAR2(50)`**	Tag	Mandatory, unique, starts with the # character
`desc_tag`	Description associated with the tag	**`VARCHAR2(255)`**	Tag	Optional
`id_image`	Unique identifier of an image	**`NUMBER(10)`**	Image	Primary key, mandatory
`desc_image`	Text description of the image	**`VARCHAR2(255)`**	Image	Optional
`chemin_fichier`	File path of the image	**`VARCHAR2(255)`**	Image	Mandatory (**`NOT NULL`**)

2. Functional Dependencies

Direct and elementary functional dependencies derived from the business rules of the *Touiteur* platform:

- **User entity:** `id_user` → `nom_user`, `prenom_user`, `email_user`, `nb_touites`
- **Tag entity:** `id_tag` → `libelle_tag`, `desc_tag`
- **Image entity:** `id_image` → `desc_image`, `chemin_fichier`
- **Touite entity:** `id_touite` → `texte_touite`, `date_pub`, `id_user`, `id_image`
- **Many-to-Many (N:M) relationships** (all-key relations with no non-key attributes): `(id_touite, id_tag)` for tags contained in a touite, `(id_suiveur, id_suivi)` for users following other users, and `(id_user, id_tag)` for user tag subscriptions.

3. Relational Schema (3NF)

Primary keys are underlined and foreign keys are prefixed with #:

- **UTILISATEUR** (**`id_user`**, `nom_user`, `prenom_user`, `email_user`, `nb_touites`)
- **TAG** (**`id_tag`**, `libelle_tag`, `desc_tag`)
- **IMAGE** (**`id_image`**, `desc_image`, `chemin_fichier`)
- **TOUITE** (**`id_touite`**, `texte_touite`, `date_pub`, `#id_user`, `#id_image`)
  Where `#id_user` (NOT NULL) references `UTILISATEUR(id_user)` and `#id_image` (NULLABLE) references `IMAGE(id_image)`.
- **CONTENIR** (**`#id_touite`**, **`#id_tag`**)
  Where `#id_touite` references `TOUITE(id_touite)` and `#id_tag` references `TAG(id_tag)`.
- **SUIVRE** (**`#id_suiveur`**, **`#id_suivi`**)
  Where both `#id_suiveur` and `#id_suivi` reference `UTILISATEUR(id_user)`.
- **ABONNER** (**`#id_user`**, **`#id_tag`**)
  Where `#id_user` references `UTILISATEUR(id_user)` and `#id_tag` references `TAG(id_tag)`.